

Prompt Engineering Mastery

Advanced techniques to get the best results from any AI

Quick Reference Cheat Sheet • AILearnHub.IO

CORE PROMPTING FRAMEWORKS

CO-STAR	Context, Objective, Style, Tone, Audience, Response. Best for structured, professional outputs.
RISEN	Role, Instructions, Steps, End-Goal, Narrowing. Best for step-by-step task completion.
RACI-P	Role, Action, Context, Input, Parameters. Best for technical and data tasks.
Chain of Thought	Instruction: "Think step by step." Forces the AI to reason through the problem before answering.
Tree of Thought	Ask AI to explore multiple reasoning paths: "Consider 3 different approaches, then pick the best."
Few-Shot	Provide 2-3 examples before your request so the AI learns the exact format you want.
Zero-Shot	No examples given — relies on the model's training. Good for simple, well-understood tasks.
ReAct	Reason + Act pattern — AI thinks, then takes an action, then observes the result. Used in AI agents.

ADVANCED PROMPT TECHNIQUES

Role Priming	"You are an expert [role] with 20 years of experience in [domain]. You always [key behaviour]..."
Constraint Setting	"Only use information from [date range/source]. Never include [exclusion]. Always include [requirement]."
Output Scaffolding	"Format your response as: [Heading] [Content] [Subheading] [Table]..." — define structure explicitly.
Iterative Refinement	Start broad, then add: "Now make it more [concise/formal/detailed]" or "Focus only on [aspect]."
Self-Consistency	Ask the same question 3 different ways. Compare answers — high agreement = more reliable output.
Persona Injection	"You are writing for [target audience]. They [know/don't know] [topic]. Adjust accordingly."
Negative Prompting	"Do NOT include [thing]. Avoid [pattern]. Never start with [phrase like 'Certainly!']"

Eval Prompting	After output: "Rate your own answer 1-10 for accuracy and rewrite if below 8."
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COMMON PROMPT MISTAKES & FIXES

Too vague	AI guesses your intent and often misses	Add role, context, format, length
No examples	AI picks a random style/format	Show 1-2 examples of ideal output
Over-constraining	AI gets confused by too many rules	Focus on 3-4 key requirements max
Single attempt	First answer is rarely the best	Always iterate: refine, challenge, expand
No format spec	AI returns unstructured walls of text	Always specify: list / table / email / JSON
Ignoring errors	AI makes mistakes you don't catch	Ask: "Check your answer for errors."

PROMPT DEBUGGING CHECKLIST

Step 1 — Role	Did you give the AI a specific expert role to play?
Step 2 — Context	Did you explain your situation / goal / audience?
Step 3 — Task	Is the task clear, specific, and achievable in one prompt?
Step 4 — Format	Did you specify the output format (list, table, email, JSON)?
Step 5 — Example	Did you show an example of what good output looks like?
Step 6 — Refine	After the first response, did you iterate and improve it?

Pro Tip: The best prompt engineers treat prompts like software — they test, iterate, and maintain a library of prompts that consistently deliver results.